

FORM VI MATHEMATICS

Time allowed: 3 hours (plus 5 minutes reading time) Exam date: 7th August 2002

Instructions:

- All questions may be attempted.
- All questions are of equal value.
- Part marks are shown in boxes in the right margin.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection:

- Each question will be collected separately.
- Start each question in a new 4-page examination booklet.
- If you use a second booklet for a question, place it inside the first. Don't staple.
- Write your candidate number on each booklet.

Checklist:

- SGS Examination booklets required — 10 booklets per boy.
- 121 boys

QUESTION ONE (Start a new examination booklet)

(a) Evaluate $\frac{2}{20 + \log_e 2}$ correct to three significant figures.

Marks
2

(b) Find the exact value of $\tan \frac{2\pi}{3}$.

1

(c) Simplify $(1 + \tan^2 \theta)$.

1

(d) Factorise completely $48x - 3x^3$.

2

(e) Find integers a and b such that $\frac{4}{2 - \sqrt{3}} = a + b\sqrt{3}$.

2

(f) Evaluate $\lim_{x \rightarrow 0} \frac{x^2 + 3x}{x}$.

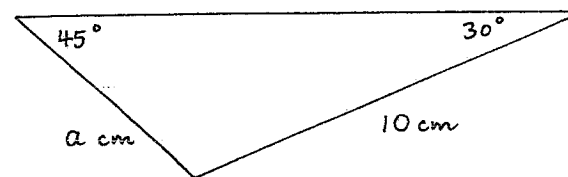
1

(g) Write down a primitive function of $\frac{1}{x+2}$.

1

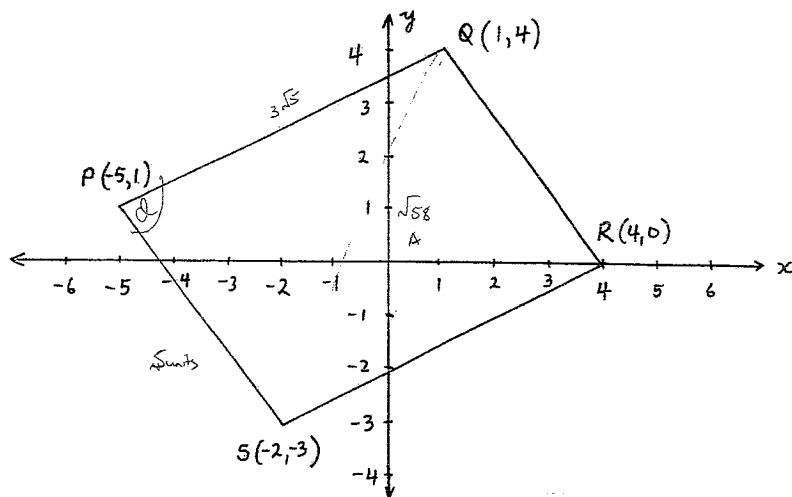
(h)

2



Find the exact value of a in the diagram above.

QUESTION TWO (Start a new examination booklet)



The quadrilateral $PQRS$ in the diagram above has vertices $P(-5, 1)$, $Q(1, 4)$, $R(4, 0)$ and $S(-2, -3)$.

- (a) Show that the length of the side PQ is $3\sqrt{5}$ units. Marks 1
- (b) Find the gradient of PQ and hence show that its equation is $x - 2y + 7 = 0$. Marks 2
- (c) Show that the perpendicular distance from S to PQ is $\frac{11}{\sqrt{5}}$ units. Marks 1
- (d) (i) Show that PR and QS have the same midpoint. Marks 2
 (ii) Hence or otherwise explain why $PQRS$ is a parallelogram. Marks 1
- (e) Find the area of the parallelogram $PQRS$. Marks 1
- (f) Given that $PS = 5$ units and $QS = \sqrt{58}$ units:
 (i) find $\angle SPQ$ correct to the nearest degree, Marks 2
 (ii) find the equation of the circle with centre $P(-5, 1)$ and radius PS . Marks 2

QUESTION THREE (Start a new examination booklet)

- (a) A parabola has equation $x^2 - 4x - 2 = -2y$. Marks 1
 (i) By completing the square, show that this equation can be written as $(x - 2)^2 = -2(y - 3)$. Marks 1
 (ii) Find the coordinates of the focus. Marks 1
 (iii) Find the coordinates of the point of intersection of the axis of symmetry and the directrix. Marks 1
- (b) Differentiate with respect to x :
 (i) $\frac{1}{2x^2}$, Marks 1
 (ii) $2x \sin x$, Marks 2
 (iii) $\frac{\log_e x}{x}$. Marks 2
- (c) Find the equation of the normal to $y = (2x - 3)^5$ at the point where $x = 1$. Marks 4

QUESTION FOUR (Start a new examination booklet)

(a) Evaluate the following definite integrals. Leave your answers in simplest form.

(i) $\int_2^4 \frac{x}{x^2 - 2} dx,$

Marks
2

(ii) $\int_0^{\frac{\pi}{2}} \sec^2 \frac{x}{3} dx.$

3

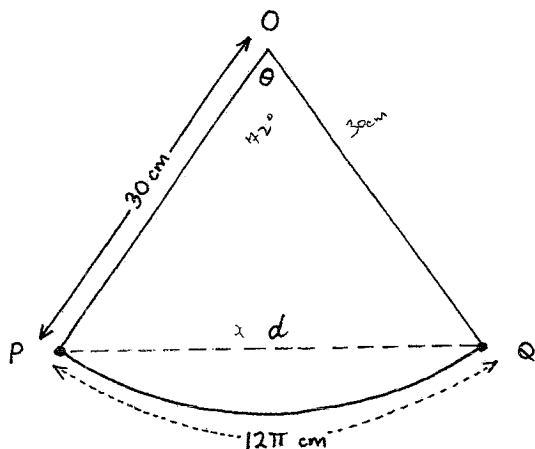
(b) (i) Sketch the graph of $y = e^{-x}$, showing the y -intercept, and shade the region bounded by the curve, the x -axis and the lines $x = 1$ and $x = -\ln 3$.

2

(ii) Find the exact area of the shaded region.

2

(c)



The diagram above shows a pendulum swinging from a fixed point O . The pendulum has length 30 cm, and the end of the pendulum swings from P to Q through an arc length of 12π cm.

(i) Show that the angle θ through which the pendulum swings is 72° .

1

(ii) Find d , the shortest distance from P to Q , correct to the nearest centimetre.

2

QUESTION FIVE (Start a new examination booklet)(a) Let α and β be the roots of the equation $2x^2 - 8x - 3 = 0$. Find:

(i) $\alpha + \beta,$

Marks
1

(ii) $\frac{1}{\alpha\beta},$

1

(iii) $\alpha^3\beta^2 + \alpha^2\beta^3.$

1

(b) Use the substitution $u = x^2 - 3x$ to solve $(x^2 - 3x)^2 - 2(x^2 - 3x) - 8 = 0$.

3

(c) During a busy cricket season, Donald increased his batting score by 4 runs in each successive innings. He was dismissed for 10 runs in his first innings of the season.

(i) How many runs did he score in his 15th innings?

1

(ii) How many innings did it take for him to score his first century of the season? (NOTE: A century is 100 runs.)

1

(iii) Show that Donald will score a total of $2n^2 + 8n$ runs in n innings.

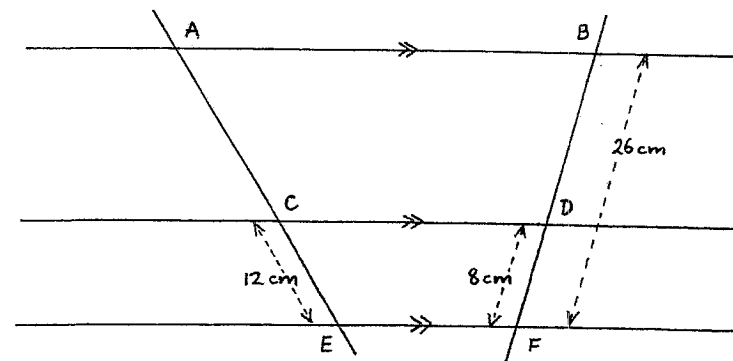
2

(iv) How many innings will it take Donald to pass 1000 runs for the season?

2

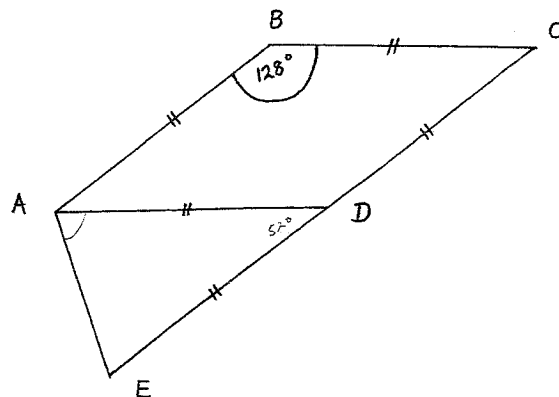
QUESTION SIX (Start a new examination booklet)

(a)

Marks
2

In the diagram above, $AB \parallel CD \parallel EF$, $BF = 26$ cm, $DF = 8$ cm and $CE = 12$ cm. Find the length of AC , stating your reason.

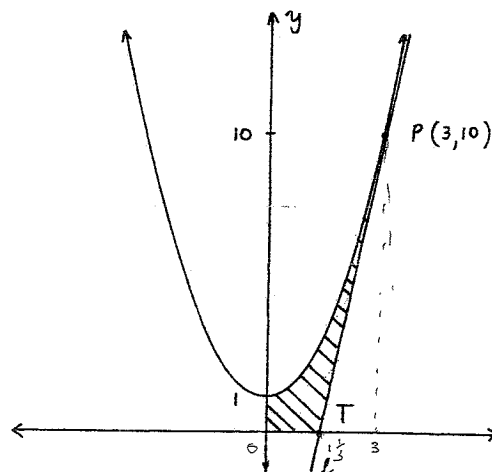
(b)



In the diagram above, $ABCD$ is a rhombus with $\angle ABC = 128^\circ$. The side CD is produced to E so that $DE = CD$. Find $\angle DAE$, giving reasons. 3

(c) Sketch $y = \frac{1}{x-3}$ showing the asymptote and the y -intercept. 2

(d)



The diagram above shows the tangent ℓ to the parabola $y = x^2 + 1$ at the point $P(3, 10)$.

(i) Show that the equation of ℓ is $y = 6x - 8$. 2

(ii) T is the point where the tangent crosses the x -axis. Show that T has coordinates $(1\frac{1}{3}, 0)$. 1

(iii) Find the area of the shaded region. 2

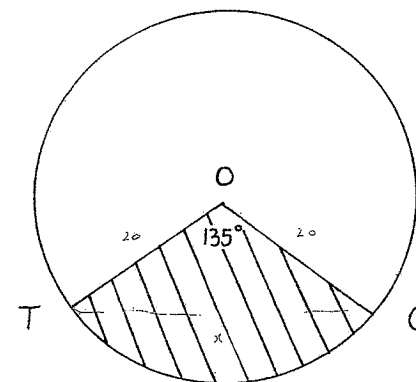
QUESTION SEVEN (Start a new examination booklet)

(a) The curve $y = f(x)$ has a gradient function $f'(x) = 3x^2 - k$, where k is a constant.

(i) Find the value of k if the curve has a stationary point at $N(-1, 3)$. 1

(ii) Hence find the equation of the curve. 2

(b)



The diagram above shows a circle with centre O . The minor sector TOC has area $150\pi \text{ cm}^2$, and $\angle TOC = 135^\circ$.

(i) Show that the circle has radius 20 cm. 2

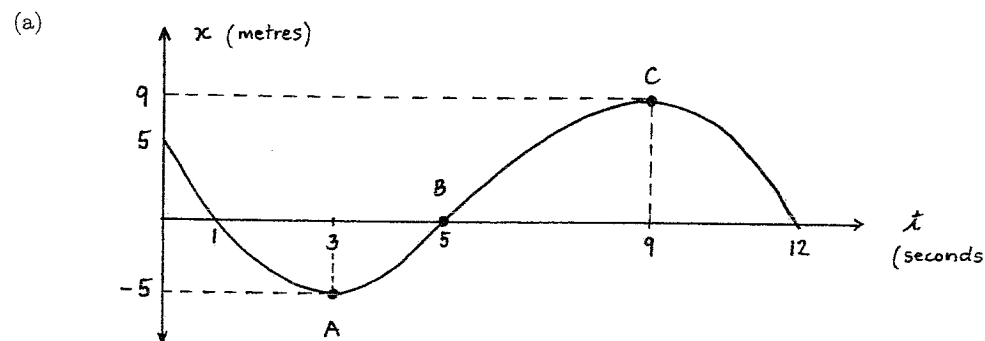
(ii) Find the area of the minor segment cut off by the chord TC , correct to the nearest square centimetre. 2

(c) Explain why the series $\cos x + \cos^2 x + \cos^3 x + \dots$ has a limiting sum when $x \neq n\pi$, for some integer n . 1

(d) (i) Sketch the graph of $y = 4 \cos 2x$ for $-\pi \leq x \leq \pi$, clearly showing the x and y -intercepts. 2

(ii) On the same set of axes, sketch the graph of $y = |x|$. 1

(iii) Hence write down the number of solutions of the equation $4 \cos 2x - |x| = 0$ for $-\pi \leq x \leq \pi$. 1

QUESTION EIGHT (Start a new examination booklet)


The diagram above shows the displacement-time graph for the first 12 seconds of a particle moving in a straight line. The points A and C are turning points and B is a point of inflexion.

- Where is the particle initially and in what direction is it travelling?
- When does the particle change direction for the first time?
- When does the maximum velocity occur?
- For what values of t is the acceleration negative?
- What is the total distance that the particle has travelled in the first 12 seconds?

Marks

2

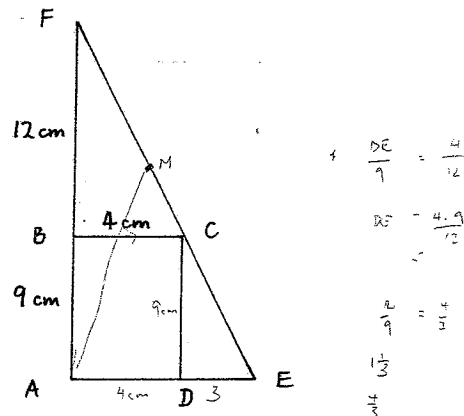
1

1

1

1

(b)



In the diagram above, ABCD is a rectangle, AB = 9 cm, BC = 4 cm and BF = 12 cm.

- Prove that $\triangle BFC \parallel \triangle DCE$, giving full reasons.
- Find the area of $\triangle AEF$.
- M lies on EF such that $AM \perp EF$. Find the area of $\triangle AME$.

2

2

2

QUESTION NINE (Start a new examination booklet)

- (a) The penguin population P on Paddy Island is decreasing according to the equation $P = Ae^{-kt}$, where A and k are constants and t is time measured in years. On 1st January 1996 there were 13 200 penguins on Paddy Island. By 1st January 2002 the penguins numbered 9900.

Marks

3

2

2

- Find the value of A and show that $k = \frac{1}{6} \ln \frac{4}{3}$.
 - If the trend continues, how many penguins will be on Paddy Island on 1st January 2010?
 - At what rate was the penguin population decreasing on 1st January 2002?
- (b) (i) Copy and complete the table below for $f(x) = (\log_e \sqrt{x})^2$, calculating each value correct to three decimal places.

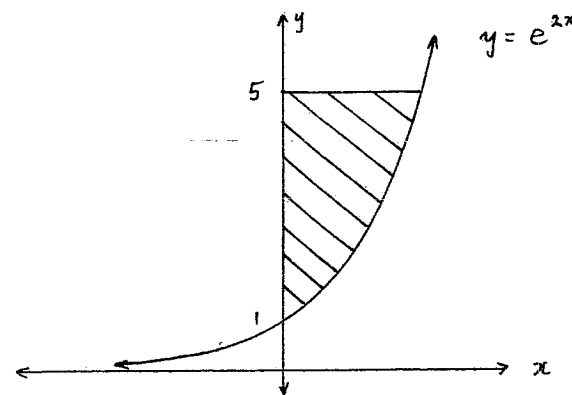
x	1	2	3	4	5
$f(x)$	0	0.120			

- (ii) Use Simpson's rule with five function values to show that

1

$$\int_1^5 (\log_e \sqrt{x})^2 dx \approx 1.22$$

(c)



The diagram above shows the region bounded by the curve $y = e^{2x}$, the y -axis and the line $y = 5$.

- Show that $x = \log_e \sqrt{y}$.
- The shaded area is rotated about the y -axis. Write down the definite integral equal to the volume formed.
- Evaluate the volume of the solid of revolution using the approximation in part(b) part(ii), leaving your answer correct to two significant figures.

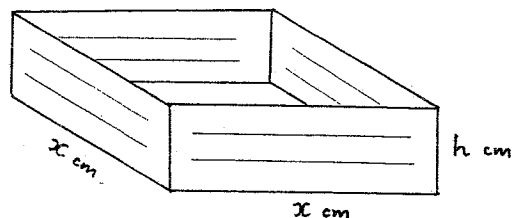
1

1

1

QUESTION TEN (Start a new examination booklet)

- (a) A metal tray, in the shape of a rectangular prism with a square base, is made out of 108 square centimetres of sheet metal. The tray is open at the top.



Let x centimetres be the side length of the base, and let h centimetres be the height.

(i) Show that $h = \frac{108 - x^2}{4x}$.

Marks

1

- (ii) Show that the volume V of the tray is given by

$$V = 27x - \frac{x^3}{4}.$$

1

- (iii) Find the maximum volume of the tray.

3

- (b) A particle moves along a straight line so that it is x metres to the right of a fixed point O at time t seconds. The acceleration of the particle is given by

$$\ddot{x} = -\frac{2\pi}{3} \sin \frac{\pi}{3}t.$$

Initially the particle is travelling with a velocity of 3 m/s.

- (i) Find the velocity v as a function of t .

1

- (ii) Find the first two times when the particle is stationary.

1

- (iii) How far does the particle travel in the first four seconds?

2

- (c) Consider the quadratic equation in x :

$$(p^2 + q^2)x^2 + 2q(p + r)x + (q^2 + r^2) = 0.$$

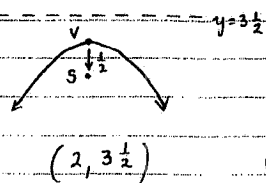
3

Find a relation, in simplest form, between p , q and r such that the quadratic has real roots.

TCW

(a) $x^2 - 4x - 2 = -2y$
 (i) $x^2 - 4x + 4 = -2y + 2 + 4$
 $(x-2)^2 = -2y + 6$
 $(x-2)^2 = -2(y-3)$ ✓

(ii) Vertex = (2, 3)
 $a = \frac{1}{2}$
 focus = (2, 2½) ✓



(iii) (2, 3½) ✓

(b) (i) $\frac{d}{dx} \left(\frac{1}{2x^2} \right) = \frac{d}{dx} \left(\frac{1}{2} x^{-2} \right)$
 $= -x^{-3}$ ✓
 $= -\frac{1}{x^3}$

(ii) $\frac{d}{dx} (2x \sin x) = 2 \sin x + 2x \cos x$ ✓✓

(iii) $\frac{d}{dx} \left(\frac{\log_e x}{x} \right) = \frac{\frac{1}{x} \cdot x - 1 \cdot \log_e x}{x^2}$ ✓
 $= \frac{1 - \log_e x}{x^2}$ ✓

(c) $y = (2x-3)^5$
 $\frac{dy}{dx} = 10(2x-3)^4$ ✓

when $x=1$, $y=-1$ ✓

$\frac{dy}{dx} = 10$

gradient of normal = $-\frac{1}{10}$ ✓

equation of normal:

$y+1 = -\frac{1}{10}(x-1)$

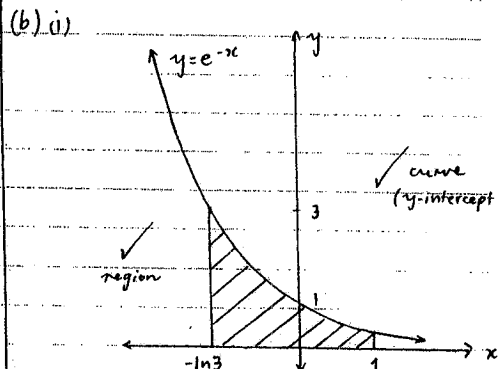
$10y+10 = -x+1$ ✓

$x+10y+9 = 0$

12

(a) (i) $\int_2^4 \frac{x}{x^2-2} dx = \frac{1}{2} \int_2^4 \frac{2x}{x^2-2} dx$
 $= \frac{1}{2} \left[\log_e(x^2-2) \right]_2^4$
 $= \frac{1}{2} (\log_e 14 - \log_e 2)$
 $= \frac{1}{2} \log_e 7$ ✓

(ii) $\int_0^{\frac{\pi}{2}} \sec^2 \frac{x}{3} dx = \left[3 \tan \frac{x}{3} \right]_0^{\frac{\pi}{2}}$ ✓
 $= 3 \tan \frac{\pi}{6} - 3 \tan 0$
 $= \frac{3}{\sqrt{3}}$ ✓
 $= \sqrt{3}$ ✓



(ii) $A = \int_{-\ln 3}^1 e^{-x} dx$ ✓
 $= \left[-e^{-x} \right]_{-\ln 3}^1$
 $= -e^{-1} + e^{\ln 3}$
 $= 3 - \frac{1}{e}$ units² ✓

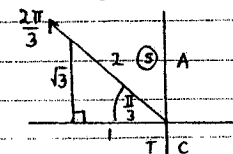
(c) (i) $\ell = r\theta$
 $\theta = \frac{12\pi}{30}$
 $= \frac{2\pi \times 180^\circ}{5}$ ✓
 $= 72^\circ$

(ii) $\sin 36^\circ = \frac{1}{2} \frac{d}{30}$ ✓
 $d = 60 \sin 36^\circ \approx 35 \text{ cm}$ ✓

12

(a) 0.0967 ✓
 (3 sig figs) ✓

(b) $\tan \frac{2\pi}{3} = -\sqrt{3}$ ✓



(c) $1 + \tan^2 \theta = \sec^2 \theta$ ✓

(d) $48x - 3x^2 = 3x(16-x^2)$ ✓
 $= 3x(4-x)(4+x)$ ✓

(e) $\frac{4}{2-\sqrt{3}} = a + b\sqrt{3}$
 LHS = $\frac{4}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}}$
 $= \frac{8+4\sqrt{3}}{4-3}$ ✓
 $= 8 + 4\sqrt{3}$

$\therefore a=8, b=4$ ✓

(f) $\lim_{x \rightarrow 0} \frac{x^2+3x}{x} = \lim_{x \rightarrow 0} \frac{x(x+3)}{x}$ ✓
 $= 3$ ✓

(g) $\log_e(x+2) + c$ ✓

(h) $\frac{a}{\sin 30^\circ} = \frac{10}{\sin 45^\circ}$ ✓
 $\frac{a}{\frac{1}{2}} = \frac{10}{\frac{1}{\sqrt{2}}}$

$a = \frac{1}{2} \times 10 \times \sqrt{2}$

$a = 5\sqrt{2}$ ✓

12

(a) $PQ^2 = (1+5)^2 + (4-1)^2$
 $= 36 + 9$
 $= 45$

$PQ = \sqrt{45}$ ✓
 $= 3\sqrt{5}$ units

(b) $m_{PQ} = \frac{4-1}{1+5}$
 $= \frac{1}{2}$ ✓

$PQ: y-1 = \frac{1}{2}(x+5)$

$2y-2 = x+5$

$x-2y+7 = 0$ ✓

(c) $d = \frac{|-2-2(-3)+7|}{\sqrt{1^2+(-2)^2}}$
 $= \frac{11}{\sqrt{5}}$ units

(d) (i) $M_{PR} = \left(\frac{-5+4}{2}, \frac{1+0}{2} \right)$
 $= \left(-\frac{1}{2}, \frac{1}{2} \right)$ ✓

$M_{QS} = \left(\frac{1-2}{2}, \frac{4-3}{2} \right)$
 $= \left(-\frac{1}{2}, \frac{1}{2} \right)$ ✓

(ii) The diagonals of PQRS bisect one another. ✓

(e) Area = bh
 $= 3\sqrt{5} \times \frac{11}{\sqrt{5}}$
 $= 33$ units² ✓

(f) (i) $\cos \angle SPQ = \frac{5^2 + (\sqrt{45})^2 - (\sqrt{58})^2}{2 \times 3\sqrt{5} \times 5}$
 $= \frac{12}{30\sqrt{5}}$
 $= \frac{2}{5\sqrt{5}}$

$\angle SPQ \approx 80^\circ$ (nearest deg)

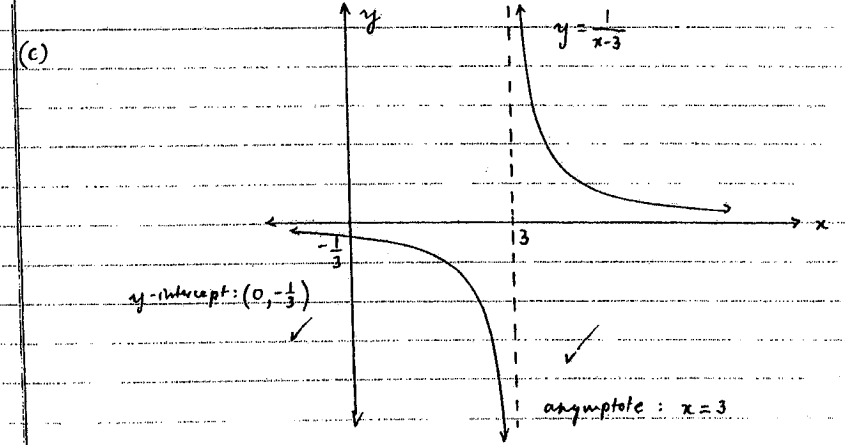
(ii) $(x+5)^2 + (y-1)^2 = 25$ ✓

12

6

(a) $\frac{AC}{CE} = \frac{BD}{DF}$ (intercepts on // lines) ✓
 $\frac{AC}{12} = \frac{18}{8}$ ✓
 $AC = 27 \text{ cm}$ ✓

(b) $\angle ADC = 128^\circ$ (opposite L's of rhombus ABCD) ✓
 $\angle DAE = \angle DEA$ (base L's of isosceles $\triangle ADE$) ✓
 $2 \angle DAE = 128^\circ$ (exterior L of $\triangle ADE$) ✓
 $\angle DAE = 64^\circ$ ✓



(d) (i) $y = x^2 + 1$
 $\frac{dy}{dx} = 2x$
 At $P(3, 10)$ $\frac{dy}{dx} = 6$ ✓

(ii) when $y=0$,
 $6x - 8 = 0$
 $6x = 8$
 $x = \frac{1}{3}$ ✓

L: $y - 10 = 6(x - 3)$
 $y - 10 = 6x - 18$
 $y = 6x - 8$ ✓

T: $(\frac{1}{3}, 0)$

(iii) Area = $\int_0^3 (x^2 + 1) dx - \frac{1}{2} \times 10 \times \frac{1}{3}$ ✓
 $= \left[\frac{x^3}{3} + x \right]_0^3 - 8\frac{1}{3}$
 $= \frac{27}{3} + 3 - 8\frac{1}{3}$
 $= 3 - \frac{2}{3} = \frac{7}{3}$ units² ✓

7

(a) $2x^2 - 8x - 3 = 0$
 (i) $\alpha + \beta = \frac{8}{2} = 4$ ✓
 (ii) $\frac{1}{\alpha\beta} = \frac{1}{-\frac{3}{2}} = -\frac{2}{3}$ ✓
 (iii) $2\alpha^3 + 2\beta^3 = 2\alpha^2(\alpha + \beta)$
 $= 2 \times \left(-\frac{3}{2}\right)^2 \times 4$
 $= \frac{9}{4} \times 4 = 9$ ✓

(b) $(x^2 - 3x)^2 - 2(x^2 - 3x) - 8 = 0$
 let $u = x^2 - 3x$
 $u^2 - 2u - 8 = 0$
 $(u - 4)(u + 2) = 0$ ✓

So $x^2 - 3x - 4 = 0$ or $x^2 - 3x + 2 = 0$
 $(x - 4)(x + 1) = 0$ $(x - 1)(x - 2) = 0$ ✓
 $x = -1, 1, 2$ or 4 ✓

(c) sequence of scores: 10, 14, 18, ... [AP: $a=10, d=4$]

(i) score in 15th innings = $a + 14d$
 $= 66$ ✓
 (ii) when $T_n = 100$, $10 + 4(n - 1) = 100$
 $4(n - 1) = 90$
 $n = 23\frac{1}{2}$

So it will take Don 24 innings to score his first cent ✓

(iii) Total runs in n innings = $\frac{n}{2}(2a + (n - 1)d)$
 $= \frac{n}{2}(20 + 4n - 4)$
 $= \frac{n}{2}(16 + 4n)$
 $= 2n^2 + 8n$

(iv) when $2n^2 + 8n = 1000$
 $n^2 + 4n - 500 = 0$
 $n = \frac{-4 \pm \sqrt{2016}}{2}$
 $n = 20.45$ (2dp)

So Don will take 21 innings to pass 1000 runs for the season.

8

- (a) (i) $x = 5$ metres, travelling in the negative direction.
 (ii) $t = 3$ seconds ✓
 (iii) $t = 5$ seconds ✓
 (iv) $5 < t < 12$ ✓ (or $5 < t < 12$, if $t=12$ is considered part of the domain)
 (v) distance = $15 + 18 = 33$ metres ✓

(b)

- (i) In Δ s BFC and DCE
 ✓ $\angle BFC = \angle DCE$ (corresponding $\angle$'s, $AF \parallel CD$)
 ✓ $\angle FBC = \angle CDE = 90^\circ$ (given ABCD is a rectangle)

$$\therefore \Delta BFC \parallel \Delta DCE \text{ (AA)}$$

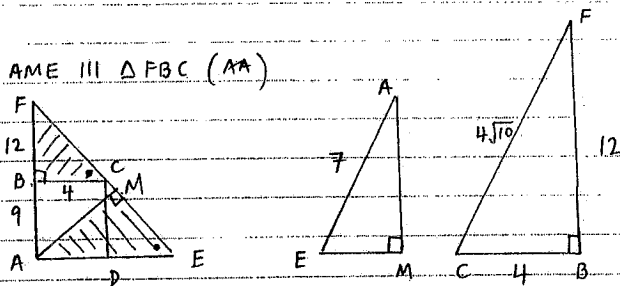
- (ii) $\frac{DE}{4} = \frac{CD}{FB}$ (matching sides of similar Δ s in the same ratio)

$$DE = 4 \times \frac{9}{12}$$

$$DE = 3 \text{ cm} \quad \checkmark$$

$$\text{Area } \Delta DEF = \frac{1}{2} \times 7 \times 21 = 73\frac{1}{2} \text{ cm}^2 \quad \checkmark$$

- (iii) $\Delta AME \parallel \Delta FBC$ (AA)



$$CF^2 = 16 + 144$$

$$CF = \sqrt{160} = 4\sqrt{10} \text{ cm}$$

$$\frac{AM}{12} = \frac{7}{4\sqrt{10}} \quad \text{(matching sides of similar } \Delta\text{s in same ratio)}$$

$$AM = \frac{21}{\sqrt{10}}$$

and

$$\frac{EM}{4} = \frac{7}{4\sqrt{10}}$$

$$EM = \frac{7}{\sqrt{10}}$$

$$\text{Area of } \Delta AME = \frac{1}{2} \times \frac{21}{\sqrt{10}} \times \frac{7}{\sqrt{10}}$$

$$= \frac{147}{20}$$

$$= 7\frac{7}{20} \text{ cm}^2 \quad \checkmark$$

12

NOTE: THERE ARE A NUMBER OF CORRECT METHODS HERE.

7

- (a) (i) $f'(x) = 3x^2 - k$
 $N(-1, 3)$ is a stationary point
 $f'(-1) = 0$
 $3 - k = 0$
 $k = 3 \quad \checkmark$

(ii) $f'(x) = 3x^2 - 3$
 $f(x) = x^3 - 3x + c \quad \checkmark$
 substitute $N(-1, 3)$: $3 = -1 + 3 + c$
 $c = 1$

$$\text{equation: } y = x^3 - 3x + 1 \quad \checkmark$$

- (b) (i) sector: $A = \frac{1}{2} r^2 \theta$
 $150\pi = \frac{1}{2} r^2 \frac{3\pi}{4} \quad \checkmark$

$$r^2 = 400$$

$$r = 20 \text{ cm} \quad \checkmark$$

- (ii) segment: $A = \frac{1}{2} r^2 (\theta - \sin \theta)$

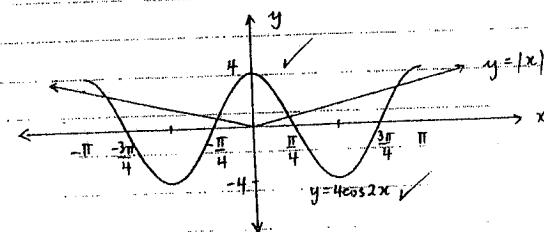
$$A = \frac{1}{2} \times 400 \left(\frac{3\pi}{4} - \sin \frac{3\pi}{4} \right) \quad \checkmark$$

$$= 150\pi - 100\sqrt{2}$$

$$\therefore 330 \text{ cm}^2 \text{ (nearest square centimetre)}$$

- (c) $\cos x + \cos^3 x + \cos^5 x + \dots$
 G.P: $r = \cos x$ and $-1 < \cos x < 1$ for $x \neq n\pi$
 so the series has a limiting sum since $-1 < r < 1$

(d) (i) (ii)



- (iii) $4 \cos 2x - |x| = 0$, $-\pi < x < \pi$, has 4 solutions.

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(10)

(a) (i) $SA = x^2 + 4xh$
 $108 = x^2 + 4xh$
 $108 - x^2 = 4xh$
 $h = \frac{108 - x^2}{4x}$ ✓

(ii) $V = l b h$
 $V = x^2 h$
 $= x^2 \left(\frac{108 - x^2}{4x} \right)$
 $= \frac{108x - x^3}{4}$ cm³

(iii) $\frac{dV}{dh} = \frac{27 - 3x^2}{4}$
 when $\frac{dV}{dh} = 0$, $\frac{3x^2}{4} = 27$
 $x^2 = 36$
 $x = 6$ ✓
 $h = \frac{108 - 36}{4}$
 $h = 3$

when $x = 6$, $\frac{d^2V}{dh^2} = -9$ ✓
 < 0 ✓
 so the volume is a maximum when $x = 6$

maximum Volume = $\frac{27(6) - 216}{4}$
 $= 108 \text{ cm}^3$ ✓

(b) (i) $\ddot{x} = -\frac{2\pi}{3} \sin \frac{\pi}{3} t$, $t \geq 0$

$v = 2 \cos \frac{\pi}{3} t + C_1$
 substitute $v = 3$ when $t = 0$:

$3 = 2 + C_1$

$C_1 = 1$

$\therefore v = 2 \cos \frac{\pi}{3} t + 1$ ✓

(ii) when $v = 0$, $2 \cos \frac{\pi}{3} t + 1 = 0$
 $\cos \frac{\pi}{3} t = -\frac{1}{2}$
 $\frac{\pi}{3} t = \frac{2\pi}{3}, \frac{4\pi}{3}, \dots$
 $t = 2, 4, \dots$ ✓

so the particle is stationary after 2 seconds then 4 seconds.

(9)

(a) (i) $P = A e^{-kt}$
 1/1/1996: when $t = 0$, $13200 = A e^0$
 $A = 13200$ ✓

1/1/2002: when $t = 6$,
 $9900 = 13200 e^{-6k}$ ✓
 $e^{-6k} = \frac{3}{4}$
 $-6k = \log_e \frac{3}{4}$
 $k = -\frac{1}{6} \log_e \frac{3}{4}$
 $k = \frac{1}{6} \log_e \frac{4}{3}$ ✓

(ii) 1/1/2010: when $t = 14$,
 $p = 13200 e^{-14k}$
 ≈ 6746 ✓

(iii) Rate of decrease: $\frac{dP}{dt} = -kP$ ✓
 1/1/2002, when $t = 6$, $\frac{dP}{dt} = -\frac{1}{6} \log_e \frac{4}{3} \times 13200 \times e^{-6k}$
 $= -\log_e \frac{4}{3} \times 2200 \times \frac{3}{4}$
 ≈ -474.68 (2dp)

so the penguin population was decreasing at a rate of approximately 475 penguins/year.

(b) (i)

x	1	2	3	4	5
f(x)	0	0.120	0.302	0.480	0.648

(ii) $\int_1^5 (\log_e \sqrt{x})^2 dx = \frac{1}{3} (0 + 4(0.120) + 0.302) + \frac{1}{3} (0.302 + 4(0.480) + 0.648)$
 ≈ 1.22

(c) (i)

$y = e^{2x}$
 $\log_e y = 2x$
 $x = \frac{1}{2} \log_e y$ ✓
 $x = \log_e \sqrt{y}$

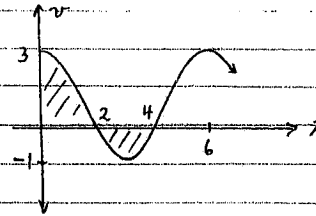
(iii) $V \approx \pi \times 1.22$
 $\approx 3.8 \text{ mm}^3$
 (2 sig figs) ✓

(ii)

$V = \pi \int_1^5 (\log_e \sqrt{y})^2 dy$ ✓

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(iii)



$$v = 2 \cos \frac{\pi}{3} t + 1$$

PERIOD = 6 s

$$\begin{aligned}
 \text{Distance} &= \int_0^2 v \, dt - \int_2^4 v \, dt \\
 &= \left[\frac{6}{\pi} \sin \frac{\pi}{3} t + t \right]_0^2 - \left[\frac{6}{\pi} \sin \frac{\pi}{3} t + t \right]_2^4 \\
 &= \frac{6}{\pi} \sin \frac{2\pi}{3} + 2 - \frac{6}{\pi} \sin \frac{4\pi}{3} - 4 + \frac{6}{\pi} \sin \frac{2\pi}{3} + 2 \\
 &= \frac{6}{\pi} \left(2 \sin \frac{2\pi}{3} - \sin \frac{4\pi}{3} \right) \\
 &= \frac{6}{\pi} \left(2 \times \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \right) \\
 &= \frac{9\sqrt{3}}{\pi} \text{ m}
 \end{aligned}$$

$$(c) \quad (p^2 + q^2)x^2 + 2q(p+r)x + (q^2 + r^2) = 0$$

For real roots $\Delta \geq 0$

$$(2q(p+r))^2 - 4(p^2 + q^2)(q^2 + r^2) \geq 0$$

$$4q^2(p^2 + 2pr + r^2) - 4(p^2q^2 + p^2r^2 + q^4 + q^2r^2) \geq 0$$

$$q^2p^2 + 2q^2pr + q^2r^2 - p^2q^2 - p^2r^2 - q^4 - q^2r^2 \geq 0$$

$$-q^4 + 2q^2pr - p^2r^2 \geq 0$$

$$q^4 - 2q^2pr + (pr)^2 \leq 0$$

$$(q^2 - pr)^2 \leq 0$$

For real roots $q^2 = pr$

(The roots are real and equal)

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TOTAL

120